LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART I

SESSION 3

Demand Forecasting & Market Sizing



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Sahay, Time Series Analysis and Forecasting (BEP8370) · Gujarati, Basic Econometrics · Sahay, Business Analytics at a Glance (BEP8363) · Facilitator: Kelechi Nnamdi

RECAP

Session 2 in Three Ideas

Shifts vs moves

Price changes move you along the demand curve; income, rivals and costs shift it.

Elasticity

Measure customers' response holding other things constant — inflation hides it.

Reference prices

Big, sudden price jumps cost more volume than the arithmetic suggests.

TODAY'S PLAN

Session 3 at a Glance — 75 Minutes

From 'how many will we sell?' to a number you can plan capacity around — and defend.



Recap & Hook

A ₦100bn plant rests on one forecast

Trend & Seasonality

Reading the shape of demand

Forecasting Honestly

Backtests, uncertainty, scenarios, and ML used responsibly

Market Sizing & Lab

Top-down, bottom-up, and the 2026 plan

BY THE END OF TODAY

Session 3 Learning Objectives

You will be able to:

01

Read a Demand Series

Separate trend, seasonality and one-off shocks in monthly sales data.

02

Forecast and Test

Build a simple trend-plus-season forecast, backtest it against rules of thumb, and express it as a range.

03

Size a Market

Estimate a Nigerian market top-down and bottom-up, and reconcile the two.

THE HOOK

A ₦100 Billion Bet on One Number

✦

“Demand grows 5% every year. Build the new line — we'll sell everything we make.”

A capacity decision is a forecast in disguise. Before approving it, ask:

↗ Is 5% the trend?

What has demand actually done over ten years, through recessions and price spikes?

▦ Which month?

Cement demand in August is far below February. Capacity must cover the peak, not the average.

⚠ How wrong could we be?

A forecast without a range is a guess with a decimal point.

NIGERIAN BUSINESS CASE

How Much Cement Will Nigeria Buy?

The situation

A Nigerian cement company is deciding whether to add a new production line for 2026. Demand has grown for a decade, but it dipped in the 2016 recession, during COVID-19 and in the 2024 price spike.

The decision question

How many tonnes of cement will Nigeria consume in **2026** — and how confident can we be?

The data we have

cement_demand_monthly.csv — 168 months (2012–2025) of national cement consumption, price per bag, a rainy-season flag and a building-permits index. *Stylised teaching data.*

What a good answer looks like

A **2026 forecast with a range and scenarios**, checked against an independent **market-sizing** estimate.

PART

01

Reading the Shape of Demand

Trend, season, shock.

THREE COMPONENTS

Every Demand Series Has Three Parts

Trend

The long-run direction: population growth, urbanisation, rising incomes. Cement: about **4.3%** a year since 2012.

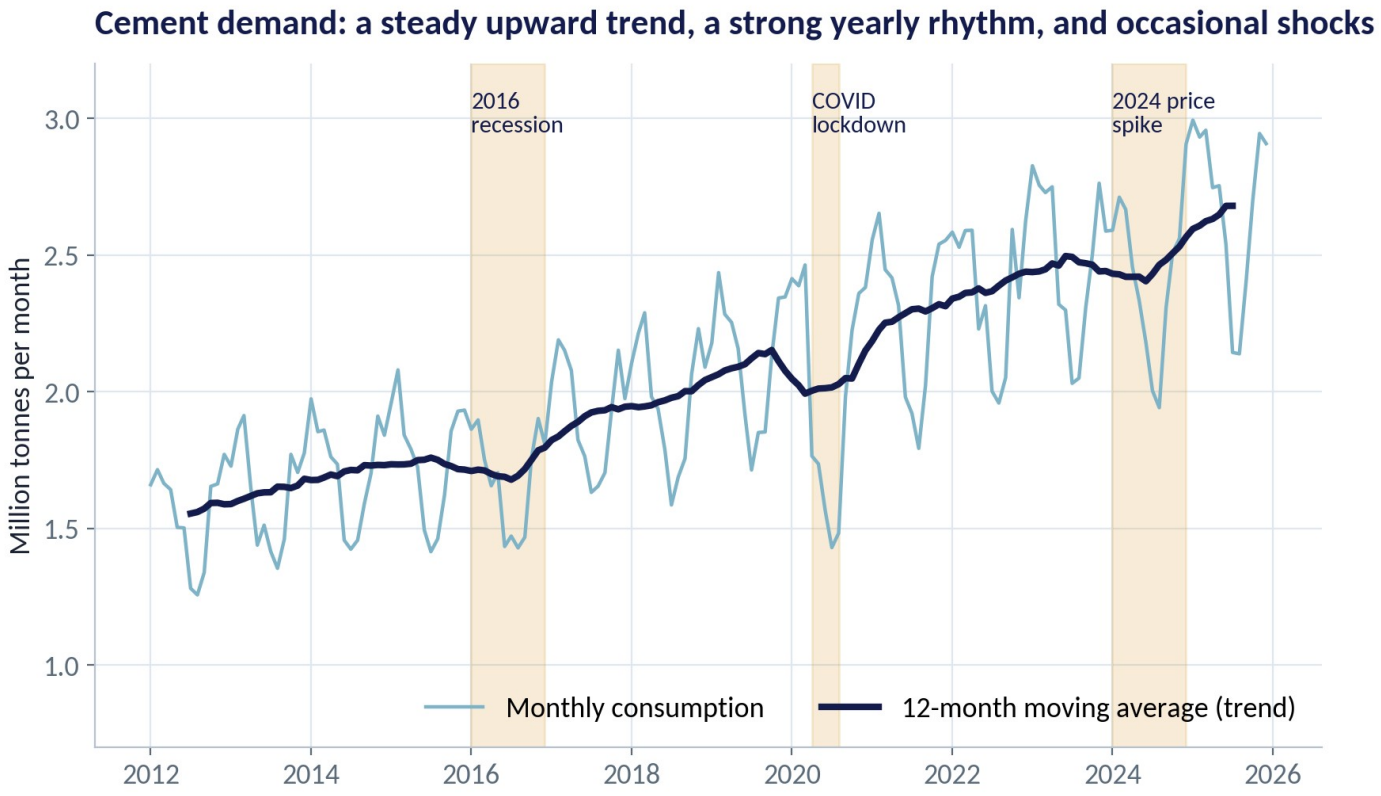
Seasonality

The pattern that repeats every year: the rainy season slows building from June to September.

Shocks & noise

Recessions, lockdowns, price spikes — and random month-to-month wobble that no model can predict.

Ten Years of Cement Demand



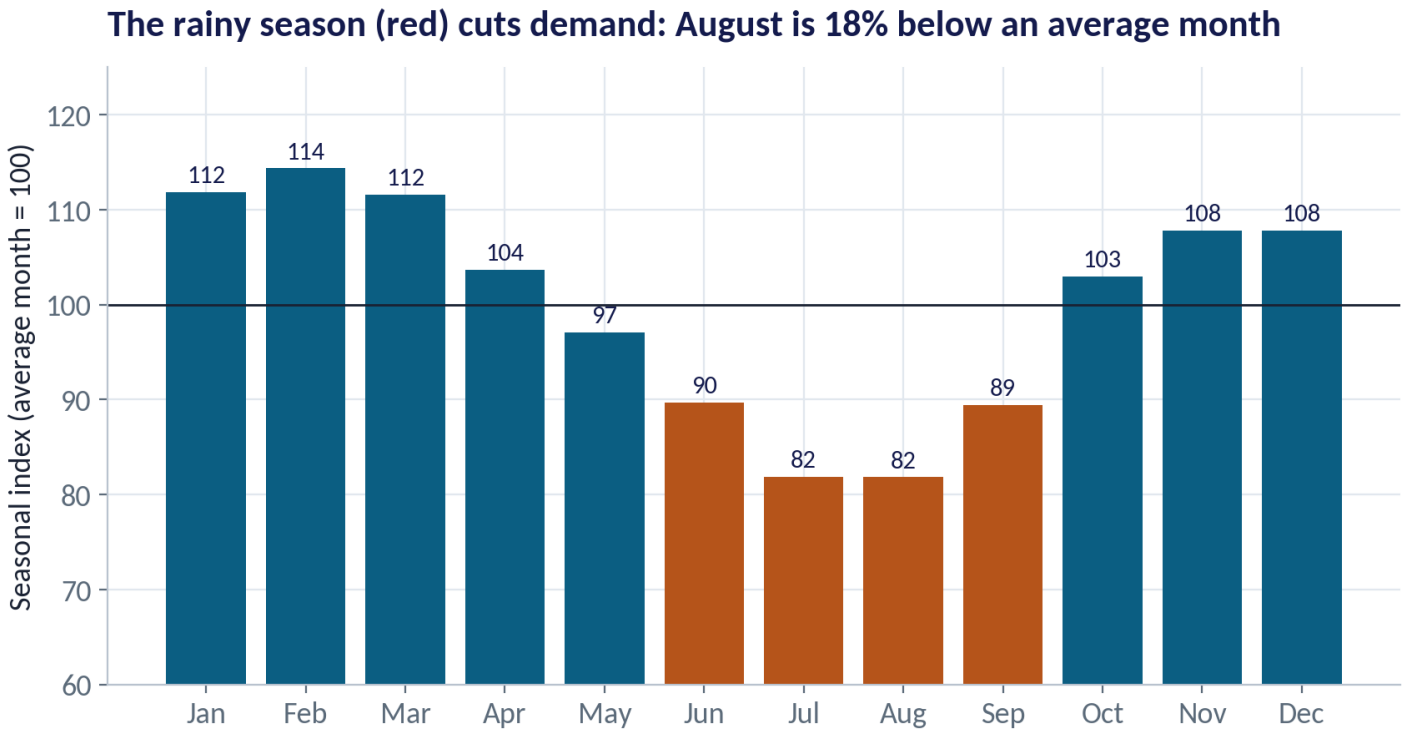
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S03-F01 · replicate: s03_figures.R / .py

WHAT TO SEE

- National consumption grew from **18.7 million tonnes** in 2012 to **32.2 million** in 2025 — about **4.3% a year**.
- The 12-month moving average smooths out the seasons and shows the **trend**.
- Three shocks: the 2016 recession, the 2020 lockdown and the 2024 price spike.
- Each time, demand recovered to the trend within about a year.

The Rainy Season Rhythm



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S03-F02 · replicate: s03_figures.R / .py

WHAT TO SEE

- February is about **14% above** an average month; July and August about **18% below**.
- Rain stops site work, so demand dips from June to September.
- Plan **capacity, stock and cash** for the peak months, not the annual average.
- Schedule maintenance shutdowns for the **rainy season**, when demand is lowest.

Trend × Season: A Forecast in One Line

IN PLAIN ENGLISH

Our forecast says: next month's demand = the **trend level** for that month × the **seasonal index** for that calendar month. We fit both from history with one regression.

EXAMPLE

Trend level for August 2026 ≈ 2.76 mt; August index ≈ 0.82 → forecast ≈ **2.26 mt**.

THE FORMULA

$$\text{forecast} = \text{trend}(t) \times \text{season}(\text{month})$$

WHY A MANAGER CARES

Simple models are **transparent**: a manager can see and challenge every assumption. That matters more than an extra decimal of accuracy.

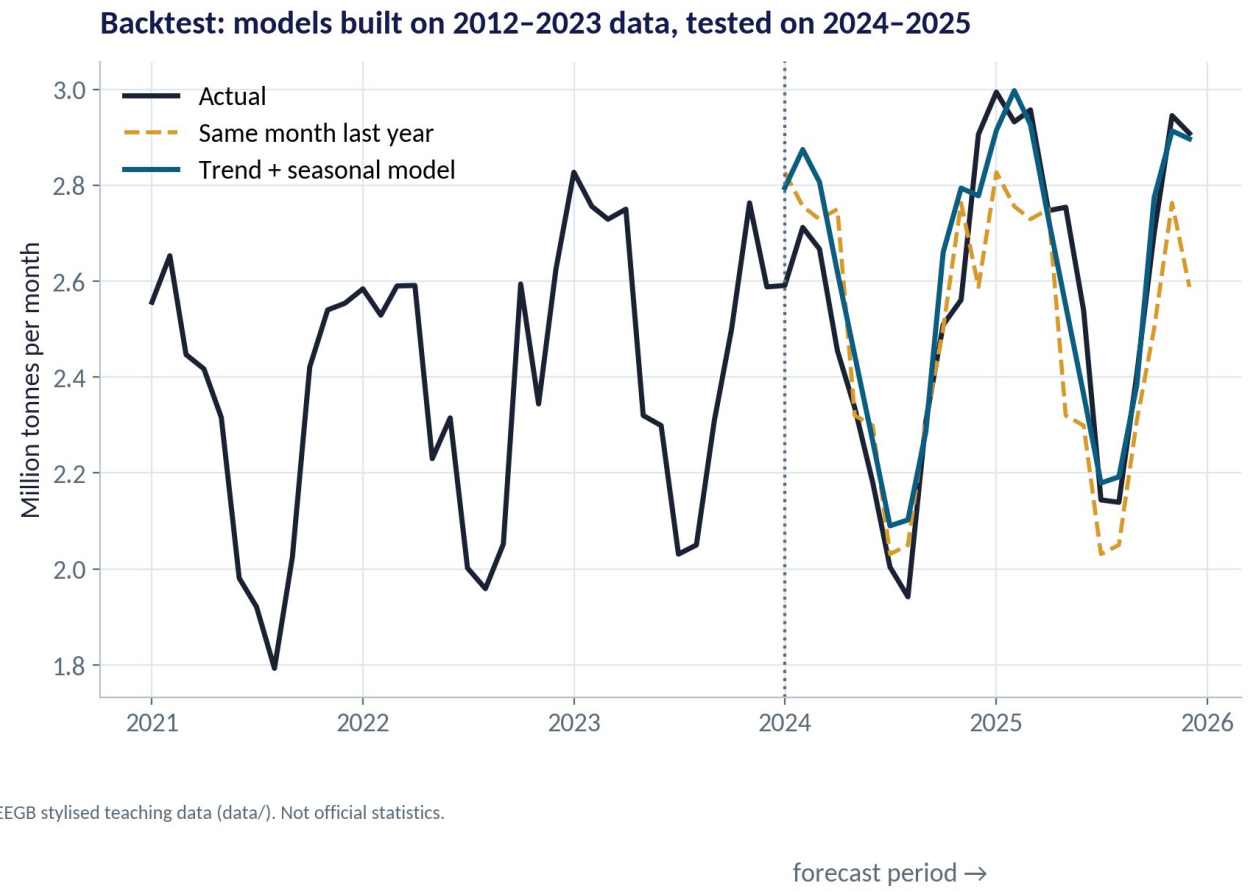
PART

02

Forecasting Honestly

Test it, give a range, plan for scenarios.

Pretend It Is January 2024



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S03-F03 · replicate: s03_figures.R / .py

WHAT TO SEE

- We fitted each method on **2012–2023 only** and asked it to forecast 2024–2025.
- Then we compared the forecasts with what actually happened.
- This is a **backtest**: the only honest way to judge a forecasting method.
- The model followed both the season and the trend; 'same month last year' missed the growth.

Which Method Forecast Best?

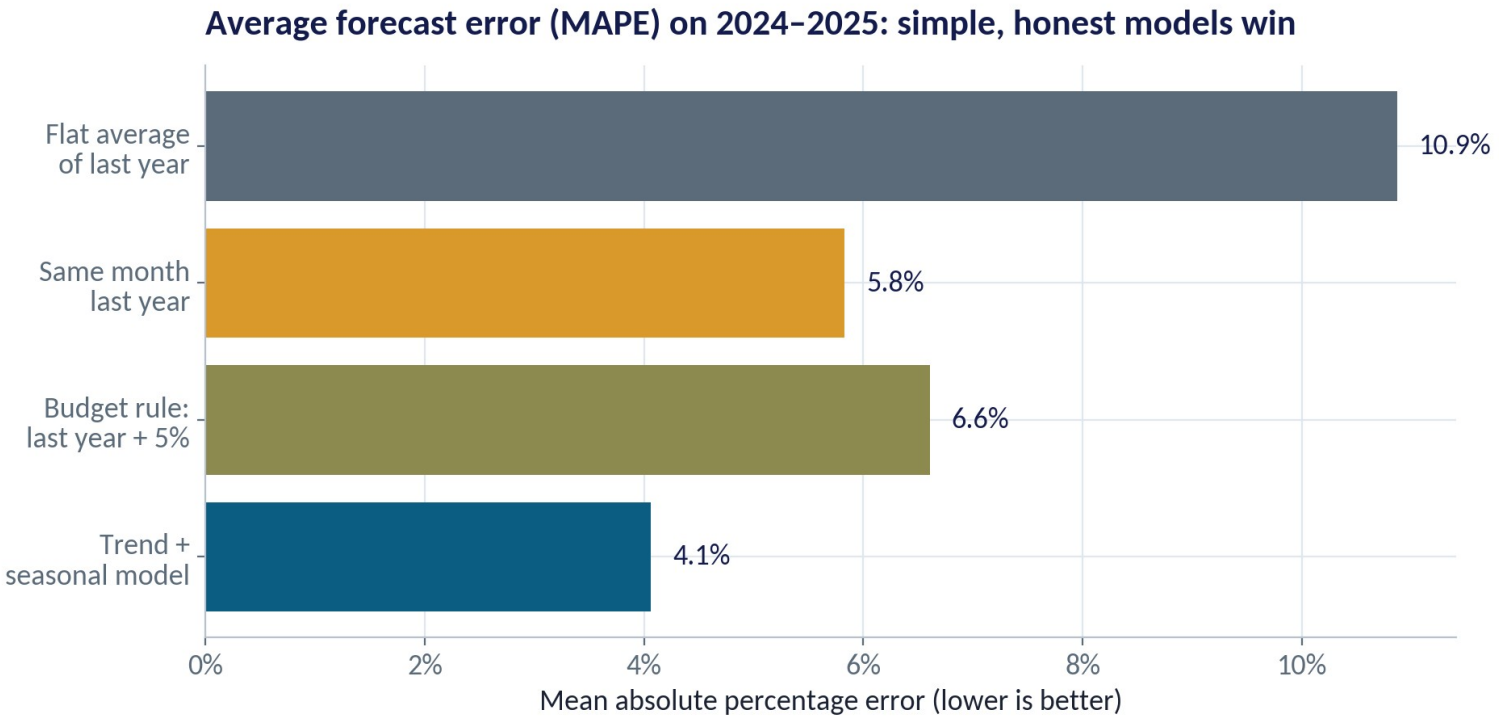


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WHAT TO SEE

- **Trend + seasonal model: 4.1%** average error — the best.
- 'Same month last year': **5.8%**. The common budget rule 'last year + 5%': **6.6%**.
- A flat average ignores the seasons: **10.9%** error.
- MAPE = mean absolute percentage error: on average, how far off were we?

TWO WAYS TO FORECAST

Judgement vs Models — Use Both

Judgemental forecasts

Sales teams, experts, management

- Know about new projects, policy changes, competitor moves
- Fast, cheap, easy to explain
- Prone to optimism and anchoring on last year
- Best for one-offs and turning points

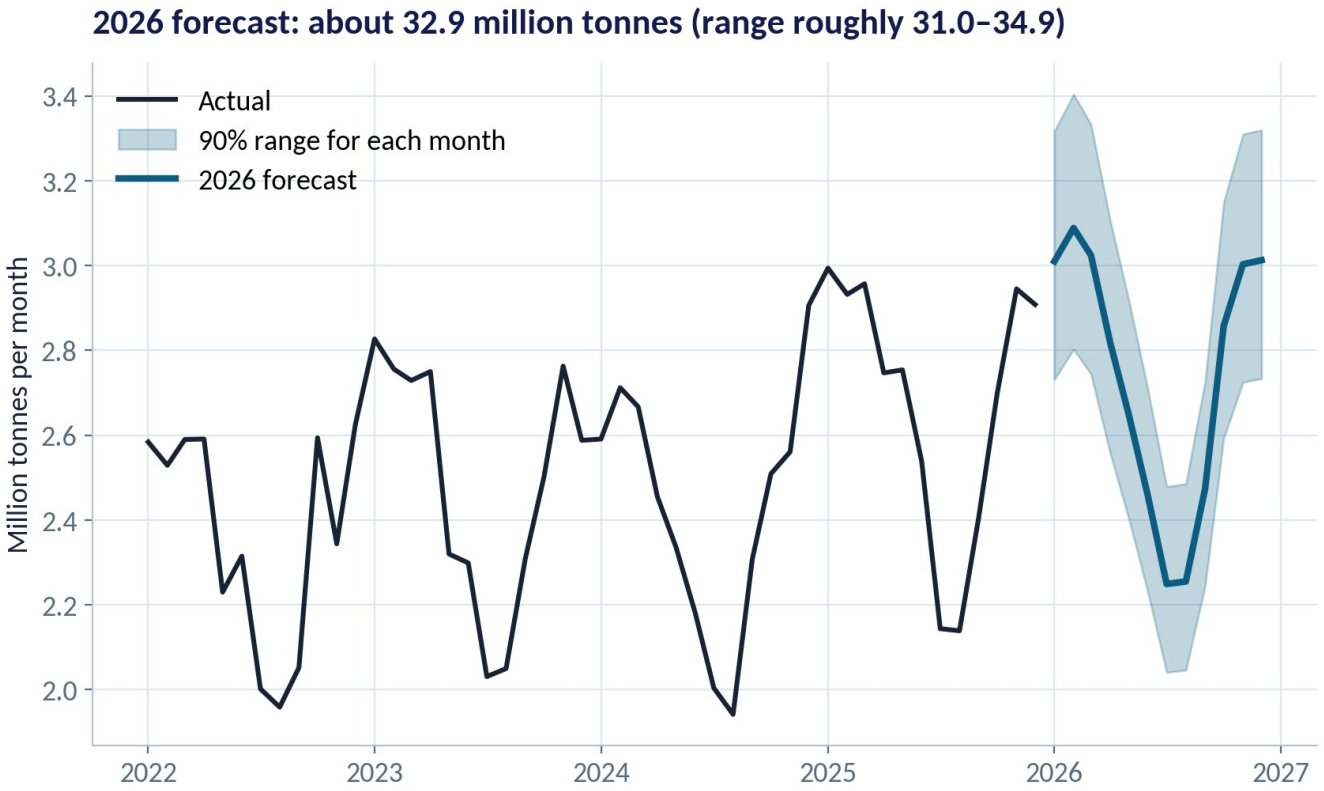
Model-based forecasts

Statistics from historical data

- Consistent and repeatable; easy to backtest
- Capture trend and season well
- Blind to events not in the data
- Best for stable, repeating patterns

*Good practice: start from the **model**, then let judgement **adjust it for named, specific reasons** — and record the adjustment so you can check it later.*

2026: A Number and a Range



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S03-F05 · replicate: s03_figures.R / .py

WHAT TO SEE

- Base forecast for 2026: about **32.9 million tonnes**.
- Monthly bands show where each month will probably fall (90% range).
- For the year: roughly **31.0–34.9 million tonnes**, based on how far past years strayed from the model.
- Say it as a range: 'about 33, very likely between 31 and 35'.

Plan for More Than One Future

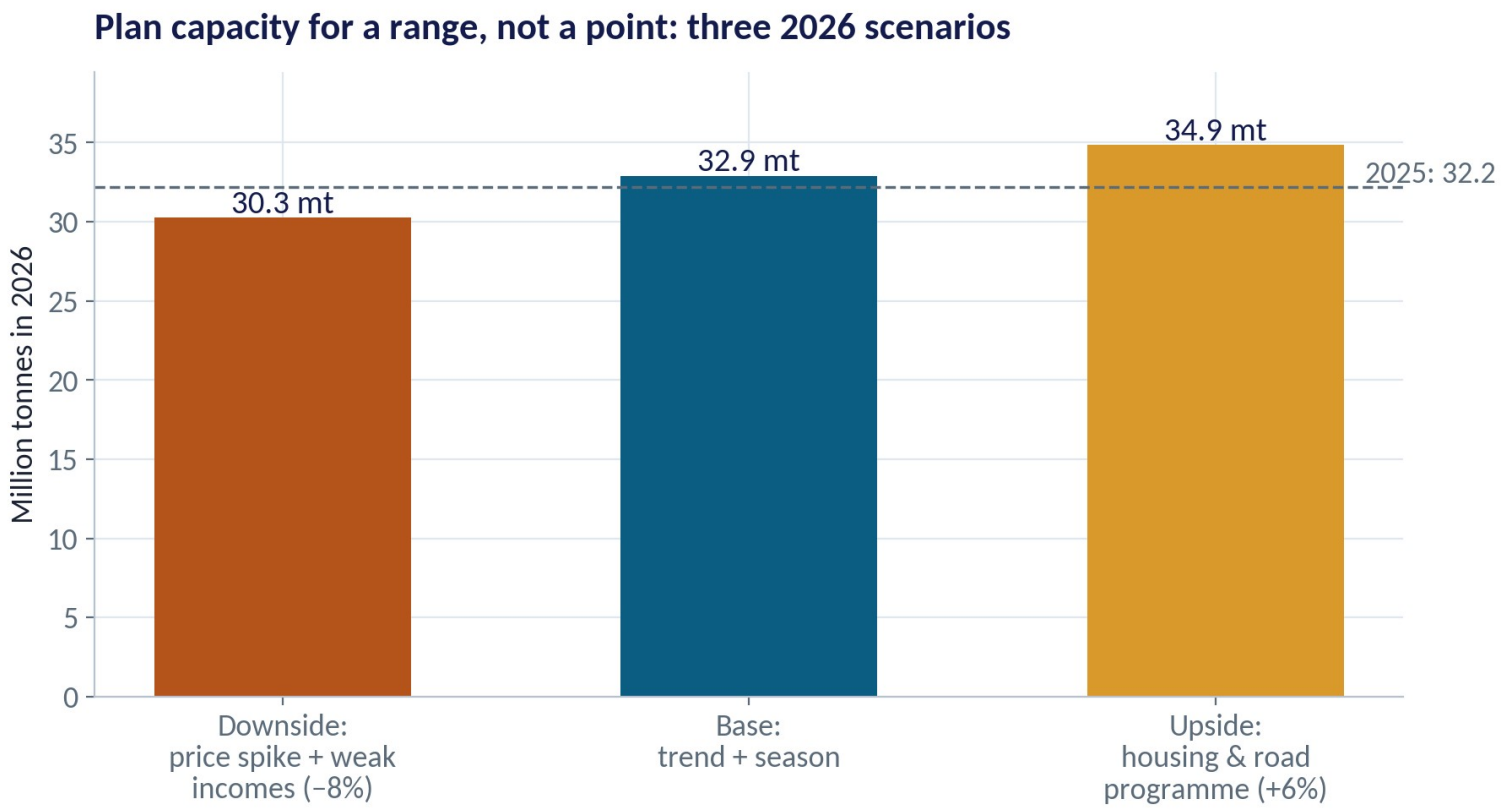


Fig S03-F06 · replicate: s03_figures.R / .py

WHAT TO SEE

- **Downside (−8%)**: another cement price spike and weak real incomes → **30.3 mt**.
- **Base**: trend and season continue → **32.9 mt**.
- **Upside (+6%)**: a large housing and road programme → **34.9 mt**.
- Ask: what capacity is **safe in the downside** and **expandable in the upside**?

More Complex Is Not More Accurate

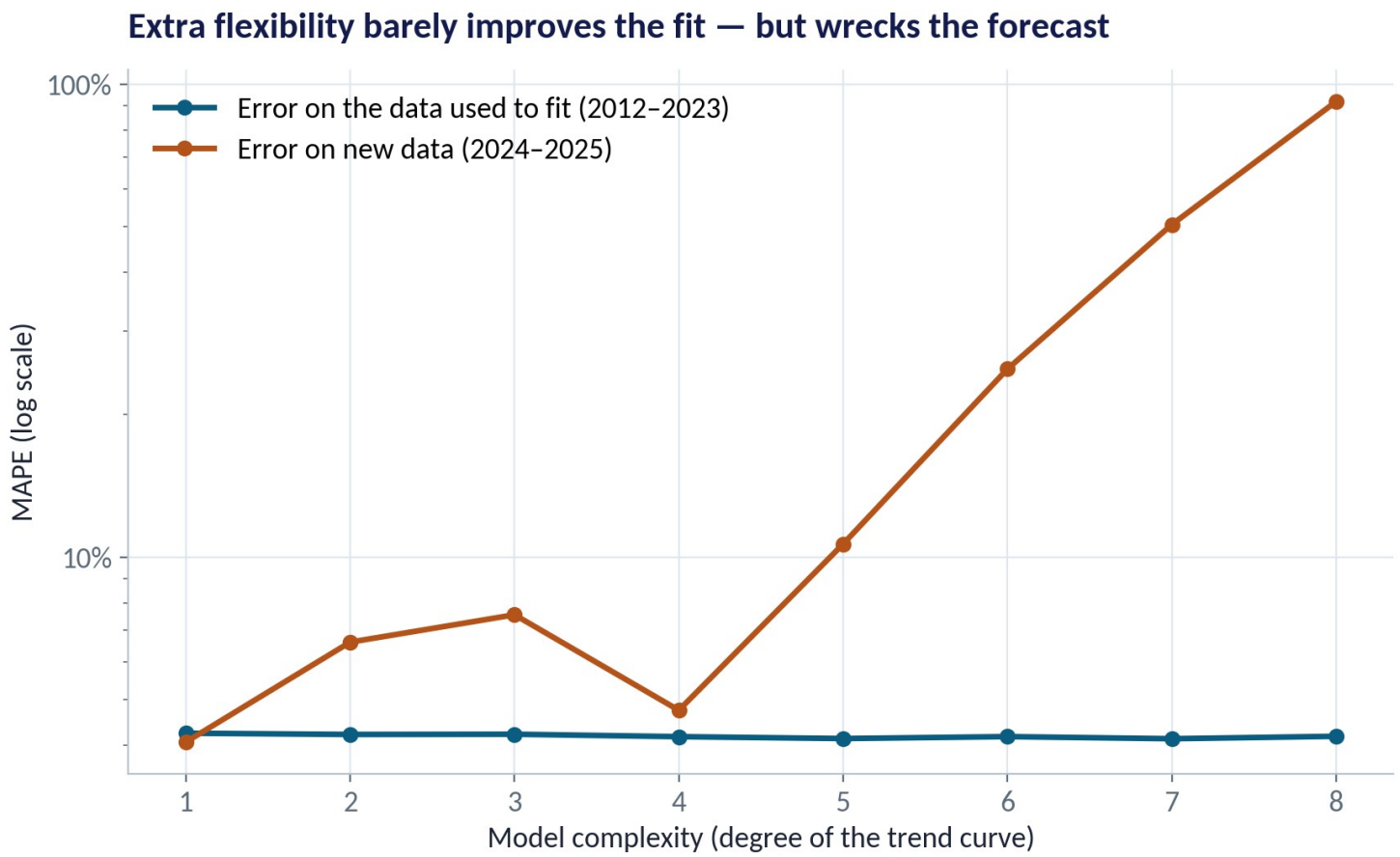


Fig S03-F08 · replicate: s03_figures.R / .py

WHAT TO SEE

- We made the trend curve more and more flexible.
- Fit to the past barely changed (about **4%** error).
- Forecasts of the future got **much worse**: from 4% to over **90%** error at the most flexible.
- This is **overfitting**: the model memorises the past instead of learning its pattern.

CONCEPT CHECK

Quick Check: What Would You Say?

?

Your sales team forecasts next year as 'this year + 10%'. A trend-plus-seasonal model forecasts +4%, with a range of −2% to +10%. What should you do?

A. Use +10% — the sales team knows the market

B. Use +4% and ignore the sales team

C. Start from the model's +4% and ask the sales team for named, specific reasons for any upward adjustment; plan capacity for the range

D. Average the two: +7%

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

03

Sizing a Market

Two routes to one number.

TOP-DOWN

Start From the Whole Economy

IN PLAIN ENGLISH

Top-down sizing starts from a big, known number — population, GDP, households — and multiplies by a rate of use per person or per naira.

EXAMPLE

230 million people × about **140 kg** of cement each per year ≈ **32.2 million tonnes**.

THE FORMULA

market = population × use per person

WHY A MANAGER CARES

Quick and easy to sanity-check against other countries. But the 'use per person' assumption carries all the weight — and it differs widely between countries at similar income levels.

Add Up the Uses

IN PLAIN ENGLISH

Bottom-up sizing builds the market from its parts: how many homes, roads, factories and repairs, times the cement each one needs.

EXAMPLE

About 450,000 new homes × ~32 tonnes = **14.4 mt**, plus repairs (6.5), public works (6.0), commercial (4.5), precast (2.4) = **33.8 mt**.

THE FORMULA

$$\text{market} = \Sigma\,(\text{units of activity} \times \text{cement per unit})$$

WHY A MANAGER CARES

Slower, but it shows **which segments drive demand** — and which ones a new government programme or a price shock would hit.

Do the Two Routes Agree?

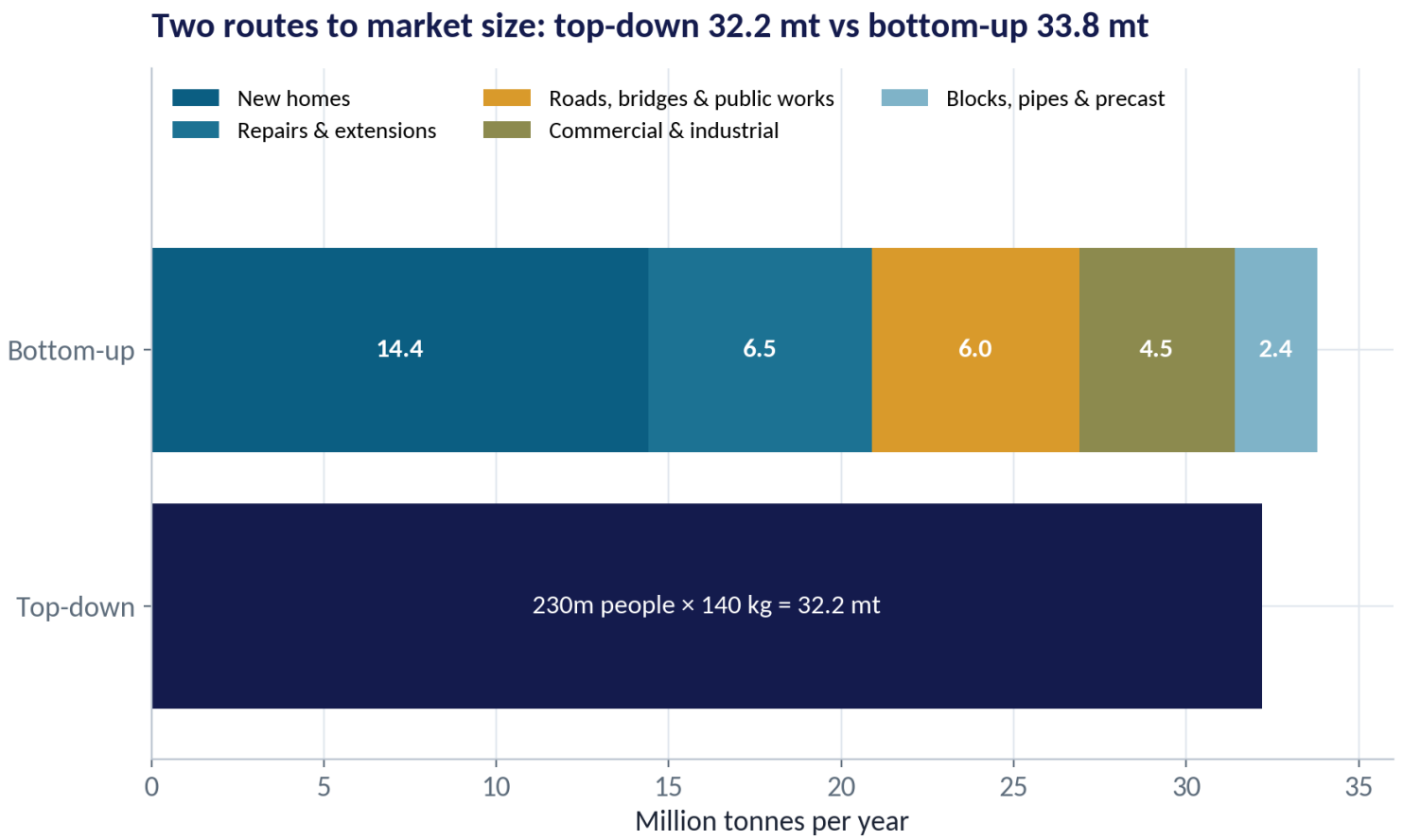


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WHAT TO SEE

- Top-down: **32.2 mt**. Bottom-up: **33.8 mt**. Within about 5% — reassuring.
- If they disagree badly, one of the assumptions is wrong. Find it before you build.
- New homes are the biggest segment: watch **mortgage rates, real incomes and land policy**.
- Both estimates sit close to the time-series forecast — three independent views pointing the same way.

Common Forecasting Mistakes

THE TRAP	THE BETTER HABIT
✗ Presenting a single number as 'the forecast'.	✓ Give a **base, a range and scenarios**, and say what would move you between them.
✗ 'Last year + X%' because that is how budgets are done.	✓ **Backtest** your method against simple rules; keep the one that wins.
✗ Planning capacity on the annual average.	✓ Plan for the **seasonal peak** and the **upside scenario**.
✗ Choosing the most complex model because it fits history best.	✓ Judge models on **data they have not seen**.
✗ Sizing the market only top-down.	✓ Cross-check with a **bottom-up** build; reconcile the two.

CONCEPT CHECK

Quick Check: What Would You Say?

?

Top-down sizing gives 32 million tonnes; bottom-up gives 45 million tonnes. What is the best next step?

A. Average them: 38.5 mt

B. Trust the bottom-up figure because it is more detailed

C. Find which assumption explains the gap (e.g. tonnes per house or homes built) and check it against independent data

D. Use whichever supports the investment case

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Cement Demand in 2026

YOU WILL NEED

Data
`data/cement_demand_monthly.csv`

R script
`sessions/s03/R/s03_lab.R`

Python script
`sessions/s03/python/s03_lab.py`

20 minutes

1. Compute **annual totals** and the average growth rate since 2012.
2. Compute the **seasonal index**: each month versus its year's average.
3. Fit a **trend + seasonal** model on 2012–2023.
4. **Backtest** it on 2024–2025 against 'same month last year' and 'last year + 5%'.
5. Forecast **2026**, and compare with the top-down and bottom-up market sizes.

Deliverable: a two-sentence 2026 capacity recommendation: how many tonnes, and what flexibility you keep.

WORDS TO OWN

Today's Vocabulary in Plain English

Trend	The long-run direction of a series.
Seasonality	A pattern that repeats at the same time every year.
Seasonal index	A month's typical level relative to an average month (100).
Moving average	An average of the last N periods; smooths out noise and seasons.
Backtest	Testing a forecasting method on past data it was not built on.
MAPE	Mean absolute percentage error: the average size of forecast misses.
Forecast range	The band within which the outcome will probably fall.
Scenario	A coherent story of the future with its own numbers.
Overfitting	A model that memorises past noise and forecasts badly.
Top-down / bottom-up	Sizing from the whole economy vs adding up the uses.

KEY TAKEAWAYS

Session 3 in 5 Ideas

01 Demand = **trend × season × shocks**; read each part separately.

02 A transparent **trend-plus-season** model beats most budget rules — **backtest** to prove it.

03 Always give a **range and scenarios**, not a single number.

04 **More complex is not more accurate**: judge models on unseen data.

05 Size markets **top-down and bottom-up**, and reconcile the two.



NEXT › Session 4: Market Structure & Platforms

Before next session: Read the Ant Financial case (917-0029-1). Come ready to say: is Ant a bank, a technology company or a marketplace — and why does the answer matter for regulators?